

Motion Detector Using NE555 Timer

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This circuit is based on a passive infrared (PIR) sensor, which automatically switches on a device when someone comes close to it. It can be used for detection of theft or an unauthorised person entering a restricted area or building. It can also turn on lights when someone approaches the area where it is installed. Applications of this circuit include security systems, corridor lights and bathroom lights, among others.

Circuit and working

The circuit diagram of the motion detector using NE555 timer is shown

in Fig. 1. It is built around 230V AC primary to 9V, 300mA secondary transformer X1, bridge rectifier DB107 (BR1), 6V voltage regulator 7806 (IC1), timer NE555 (IC2) and a few other components.

The 230V AC mains is stepped down to 9V, 300mA through step-down transformer X1. Secondary output of X1 goes to bridge rectifier BR1. Capacitor C1 filters the ripples and the rectified output is given to regulator 7806 (IC1). IC1 provides 6V regulated DC output to operate the circuit. LED1 is used as a power-on indicator.

IC2 is configured in monostable mode. Time period of IC2 is based on resistor R4 and capacitor C3, which is

around ten minutes in this case. By changing resistor R4 and capacitor C3 you can change the time period of IC2.

Output pin 3 of IC2 is connected to the base of relay driver transistor T2. Contacts of relay RL1 are connected to the load, which could be a CFL or a bulb connected across CON2.

The PIR sensor is a pyroelectric device developed for detection of hu-

Test Points

Test point	Details
TP0	0V (GND)
TP1	6V
TP2	Normally high and becomes low when motion is detected

PARTS LIST

Semiconductors:

- IC1 - 7806, 6V voltage regulator
- IC2 - NE555 timer
- BR1 - DB107 bridge rectifier
- LED1 - 5mm LED
- D1 - 1N4007 rectifier diode
- T1, T2 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 680-ohm
- R2 - 10-kilo-ohm
- R3, R5 - 1-kilo-ohm
- R4 - 560-kilo-ohm

Capacitors:

- C1 - 220 μ F, 25V electrolytic
- C2 - 0.1 μ F ceramic disk
- C3 - 1000 μ F, 16V electrolytic
- C4 - 0.01 μ F ceramic disk

Miscellaneous:

- CON1 - 3-pin connector
- CON2-CON4 - 2-pin connector terminal
- X1 - 230V AC primary to 9V, 300mA secondary transformer
- RL1 - 6V, 1C/O relay
- Load - 40W, 230V AC bulb
- PIR sensor

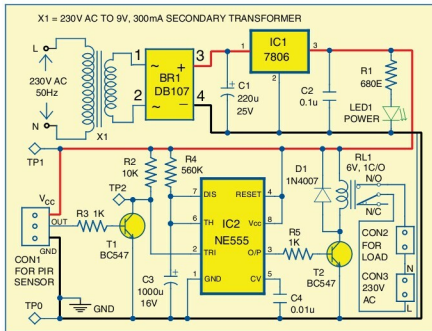


Fig. 1: Circuit diagram of the motion detector



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